



SULLAMUSSALAM SCIENCE COLLEGE, AREEKODE

**INTERNSHIP REPORT ON
FIBER OPTIC COMMUNICATIONS,
FTTH NETWORK ARCHITECTURE
AND ELECTRICAL HOME APPLIANCES MAINTENANCE**

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Submitted in partial fulfillment of the BSc Physics Degree

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**In Industrial Collaboration with:
Bfone Fibernet, Edavannappara
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CERTIFICATE

This is to certify that the internship report entitled “Fiber Optic Communications, FTTH Network Architecture and Electrical Home Appliances Maintenance” is a bona fide record of the industrial training carried out by **SALMAN FARIS** under my academic supervision and guidance, in corporate collaboration with Bfone Fibernet, Edavannappara, in partial fulfillment of the requirements for

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Internship Activity Schedule

The day-wise internship activities and topics covered during the training period are summarized in the table below:

Day	Date	Topics / Activities Covered
Day 1	21.05.2026	Optical Fiber Theory Class (Introduction to OFC)
Day 2	22.05.2026	Splicing Practical Section
Day 3	23.05.2026	Repairing of the Ceiling Fan
Day 4	25.05.2026	• Theory Class of Fiber To The Home (FTTH) • Repairing of Mixer Grinder • Field Visit of Splicing
Day 5	26.05.2026	Field Visit at Manjeri Telephone Exchange
Day 6	08.06.2026	Power Calculation Theory Class
Day 7	09.06.2026	• Repairing of Iron Box • Repairing of Wall Fan • Repairing of Induction Cooker

Table 1: Day-wise Internship Activity Log

Part I

Fiber Optic Communications and FTTH Network Architecture

Part Introduction: Fiber Optic Infrastructure

Modern communications are fundamentally driven by high-speed data transfer infrastructures operating on optical physics principles. Moving beyond legacy copper-wire networks, optical fibers guide light signals to transport large amounts of data over vast distances with minimal signal degradation. This section of the report details a technical analysis of modern Fiber-Optic Communication (OFC) and Fiber-to-the-Home (FTTH) setups, based on practical experience at Bfone Fibernet. The primary objective of this module is to comprehensively trace the optical signal pathway from its structural origins to real-world residential deployment. These chapters break down the principles of Total Internal Reflection (TIR) in fiber cores, the anatomy of optical cables, and mathematical modeling for link loss budgeting. Furthermore, this section details practical operational protocols, including high-precision fusion splicing, color-coding standards, network diagnostics, and troubleshooting methodologies implemented in active field environments. This section finishes with a technical summary of our field study at the Manjeri BSNL Exchange Office.

Chapter 1

Introduction to Optical Fiber Communication

1.1 Overview

Optical Fiber Communication (OFC) refers to the technique of transferring information across distances using modulated light pulses guided through thin fibers. In this system, light acts as an electromagnetic carrier wave modulated to carry digital and analog data.

1.2 The Basic Fiber Optic Link

A standard fiber-optic link is made up of a sequence of core hardware blocks:

- Encoder/Source: Converts data into electrical signals.
- Transmitter (E/O): Converts electrical signals into optical signals using LEDs or LASERs.
- Optical Fiber: The medium that guides the light signal.
- Receiver (O/E): Uses a photo-detector (PIN or APD) to convert light back into electrical signals.
- Decoder: Recovers the original data from the signal.

1.3 Advantages of Fiber Optics

- Non-conductive: No risk of sparks or electrical hazards.
- Electromagnetic Immunity: Immune to EMI and lightning.
- Large Bandwidth: Can carry massive amounts of data.
- Low Loss: Signal attenuation is significantly lower than copper cables.
- Small and Lightweight: Easier to install and transport.

Chapter 2

Principle and Cable Structure

2.1 Principle of Operation: Total Internal Reflection

Light is guided along the length of the optical fiber core using the principle of Total Internal Reflection (TIR). The occurrence of TIR requires two strict conditions:

- The refractive index of the core (n_1) must be greater than the cladding (n_2).
- The angle of incidence must be greater than the critical angle (θ_c).

The critical angle value is determined by:

$$\theta_c = \sin^{-1} \left(\frac{n_2}{n_1} \right) \quad (2.1)$$

2.2 Cable Anatomy

The physical architecture of a standard fiber cable consists of these core elements:

- Core: The innermost glass or plastic part where light travels.
- Cladding: Layer with a lower refractive index that reflects light back into the core.
- Buffer: Protects the fiber from moisture.
- Strength Member: Usually Aramid yarn (Kevlar) to prevent cable breakage.
- Outer Jacket: Protects against environmental damage.

2.3 Cable Types

- Tight-Buffer Cable: Used mainly for indoor applications and patch cords.
- Loose-Tube Cable: Used for outdoor environments (ducts, direct burial).
- Ribbon Cable: Dozens of fibers packed together for high-density applications.

Chapter 3

Link Budgeting, Splicing, and Mathematical Calculations

3.1 OFC Splicing

Splicing is used to make permanent connections between fiber ends. The two main types are:

- Fusion Splicing: Using an electric arc to melt and fuse fibers.
- Mechanical Splicing: Aligning fibers using a mechanical fixture.

3.2 Logarithmic Conversion Mechanics

To make loss calculations additive rather than multiplicative, optical power is measured using the logarithmic decibel-milliwatt (dBm) scale:

$$P_{\text{dBm}} = 10 \log_{10} \left(\frac{P_{\text{mW}}}{1 \text{ mW}} \right) \quad (3.1)$$

3.3 Splitter Loss Calculations

An optical splitter's theoretical insertion loss increases mathematically with the number of output ports (N):

$$\text{Loss}_{\text{Ideal}} = 10 \log_{10}(N) \quad (3.2)$$

3.4 Standard Loss Values for Link Performance

For network design and planning, the standard component attenuation values are defined as:

Network Element Component	Standard Loss Value
Fiber Path Loss @ 1310 nm / 1490 nm	0.40 dB per km
Fiber Path Loss @ 1550 nm	0.25 dB per km
Fusion Splice Junction	0.10 dB per splice
SC/UPC Physical Connector Pair	0.35 dB per connector
SC/APC Physical Connector Pair	0.25 dB per connector
1 × 2 Splitter Module	3.0 dB average insertion loss
1 × 16 Splitter Module	13.0 dB average insertion loss
1 × 32 Splitter Module	16.0 dB average insertion loss

Table 3.1: Standard Loss Parameters Used in FTTH Networks

3.5 Mathematical Verification: Power Link Budget Case Study

$$P_r = P_t - \text{Loss}_{\text{total}} \quad (3.3)$$

3.5.1 Case Study Parameter Blueprint

As a practical case study, consider a 7 km fiber link installed by Bfone Fibernet. The link features 4 fusion junctions, 2 SC/UPC connector pairs, and a cascaded splitter configuration consisting of a 1 × 16 and a 1 × 8 unit. The loss budget details for this specific fiber span are calculated below:

- Fiber Line Attenuation Loss: $7 \text{ km} \times 0.40 \text{ dB/km} = 2.8 \text{ dB}$
- First Stage Insertion Loss (1 × 16 Splitter) = 13.0 dB
- Second Stage Insertion Loss (1 × 8 Splitter) = 10.0 dB
- Total Mechanical Splice Losses: $4 \times 0.10 \text{ dB} = 0.4 \text{ dB}$
- Total Dynamic Connector Losses: $2 \times 0.35 \text{ dB} = 0.7 \text{ dB}$

The sum of these individual components represents the total link attenuation:

$$\begin{aligned} \text{Total Attenuation Loss} &= 2.8 + 13.0 + 10.0 + 0.4 + 0.7 \\ &= 26.9 \text{ dB} \end{aligned}$$

3.5.2 Link Status Analysis

The acceptable operating window for standard ONT fiber modems is between -10 dBm and -28 dBm: Minimum Receiver Threshold $\geq -28 \text{ dBm}$

$$\text{Receiver Sensitivity Threshold} \leq -28 \text{ dBm} \quad (3.4)$$

With a calculated link loss of 26.9 dB and a transmitter power of 0 dBm, the signal strength at the customer premises is -26.9 dBm. Because this value is stronger than the -28 dBm sensitivity limit, the link will operate successfully.

Chapter 4

Optical Fiber Splicing: Detailed Analysis

4.1 What is Splicing?

Splicing refers to the permanent joining of two optical fiber cores to allow continuous light propagation. While connectors are built for quick disconnects, splices are permanent joints optimized for stability. The main objective of splicing is to minimize insertion loss and keep return loss (back-reflection) low.

4.2 Types of Splicing

Technicians utilize two primary methods for joining fibers:

4.2.1 Fusion Splicing

This is the premium industry-standard method, which utilizes an electric arc to melt the glass tips and fuse them permanently. Attenuation: Usually between 0.01 dB and 0.05 dB. Applications: Used in long-distance backbones and residential FTTH deployments.

4.2.2 Mechanical Splicing

This technique uses a mechanical alignment sleeve containing index-matching gel to hold the fibers in alignment. Attenuation: Usually ranges from 0.2 dB to 0.5 dB. Applications: Temporary fiber joins, emergency field repairs, and low-cost terminations.

4.3 Importance and Significance

- **Link Extension:** Connecting multiple segments to build long-distance networks.
- **Signal Integrity:** Maintaining near-perfect light transmission with minimal attenuation.
- **Back-reflection Control:** Minimizing return loss, which protects laser sources.
- **Durability:** Providing maintenance-free connections that withstand environmental stress.

4.4 Step-by-Step Fusion Splicing Process

1. Stripping: The 250 μm outer coating is removed using specialized fiber strippers without scratching the glass.
2. Cleaning: The bare glass fiber is thoroughly cleaned using 99% Isopropyl Alcohol (IPA) and lint-free wipes to remove all contaminants.
3. Cleaving: The fiber is cut using a precision diamond-blade cleaver to ensure a perfect 90-degree end face (angle $\leq 0.5^\circ$).
4. Fusion: The fiber ends are aligned using the Profile Alignment System (PAS) and fused using an electric arc at 2000°C.
5. Protection: A heat-shrink splice sleeve is applied and heated to protect the joint from physical damage and moisture.

4.5 Advantages

- Lowest possible insertion loss.
- Permanent and highly reliable joints.
- Resistant to vibration and temperature fluctuations.
- Space-efficient for high-density fiber enclosures.

4.6 Future Scope

- Automation: Developing fully autonomous AI-driven splicing robots for sub-sea and hazardous environments.
- Multi-core Fiber Splicing: Moving beyond single-core to support high-density multi-core fibers to increase bandwidth capacity.
- Enhanced Precision: Integrating advanced nanotechnology for near-zero loss splicing in extreme environments.

4.7 Conclusion

Ultimately, fiber splicing is a core skill in optical telecommunications. Permanent joins minimize attenuation, allowing light signals to traverse long distances. For primary network segments, fusion splicing is dominant, whereas mechanical splicing remains useful for emergency restoration work.

Chapter 5

Introduction to FTTH & Passive Optical Networks (PON)

5.1 Overview of FTTH

FTTH represents the transition to complete optical networking, replacing copper phone networks (DSL) and coax networks with dedicated optical fibers directly to the home. The dominant technology in FTTH is PON, or Passive Optical Network. It is 'passive' because there are no powered switches in the field; light is divided using glass splitters. This design reduces power consumption, cuts down on hardware failures, and lowers operating costs. Through this point-to-multipoint topology, a single fiber feeds multiple households.

5.2 The FTTH Network Architecture Layers

An FTTH network's physical layout is categorized into three key segments: A. The Feeder Segment: Starts at the service provider's Central Office (CO) and runs to a local neighborhood hub known as the Fiber Distribution Hub (FDH) or Primary Cross-Connect box. This segment uses high-core-count cables (often containing 48 to 144+ fibers) designed to carry aggregated data over long distances (up to 20 kilometers) without splitting.

B. The Distribution Segment: Travels from the FDH to smaller street-level ODP (Optical Distribution Point) or FAT (Fiber Access Terminal) boxes. At the FDH, the feeder fibers are connected to splitters, and the split fibers branch out into the neighborhood using medium-core cables (12 to 24 fibers) along utility poles or underground conduits.

C. The Drop Segment: The final link that bridges the ODP/FAT on the street directly into the customer's living room or office. This utilizes a Drop Cable, which is a highly specialized, lightweight, ruggedized fiber cable (usually containing only 1 or 2 fibers) engineered to survive tight bends and physical pulling.

5.3 Core Hardware Components

FTTH operations depend on a clear split between active and passive network elements:

5.3.1 Active Components (Power Required)

OLT (Optical Line Terminal): Installed at the Central Office (CO). It acts as the central 'brain' of the network, converting digital IP data from the backbone into laser pulses and scheduling traffic for thousands of subscribers.

ONT / ONU (Optical Network Terminal / Unit): Installed inside the subscriber's home. It acts as the translator, receiving optical light signals and converting them back into standard electrical signals (Ethernet/Wi-Fi) for consumer devices.

5.3.2 Passive Components (No Power Required)

Optical Splitters: The heart of PON. A passive glass prism component that takes a single incoming beam of light and splits it into multiple identical streams to serve multiple users. Standard split ratios are 1:8, 1:16, 1:32, or 1:64.

ODF / FDH Hubs: Environmental enclosures or cabinets that house splices and splitters, protecting connections from water, dust, and temperature changes.

Chapter 6

FTTH Deployment, Coding and Testing

6.1 Fiber Color Coding Standard (TIA-598-C)

To manage multi-core cables without confusion, a standard 12-color sequence is followed. When core counts exceed 12, the color code is repeated with a colored tracer thread.

No.	Fiber Color	Symbol	No.	Fiber Color	Symbol
1	Blue	BL	7	Red	RD
2	Orange	OR	8	Black	BK
3	Green	GR	9	Yellow	YL
4	Brown	BR	10	Violet	VI
5	Slate (Grey)	SL	11	Rose (Pink)	RS
6	White	WH	12	Aqua	AQ

Table 6.1: TIA-598-C Color Coding Identification

6.2 Physical Deployment & Site Survey

6.2.1 Site Survey & Planning

The deployment process begins with a detailed site survey comprising:

- Route Mapping: Finding the shortest, most efficient path from street FAT to house.
- Structural Assessment: Identifying building entry points and optimal indoor ONT placement.
- Safety Check: Keeping clear of electrical mainlines to avoid physical interference.
- BOM Calculation: Generating the exact Bill of Materials (cables, clamps, rosettes).

6.2.2 Outdoor Routing Operations

Aerial Deployment: Fiber cables are strung between utility poles using tension/anchor clamps. Slack loops must be left at ends to accommodate weather-induced physical shifts. Underground Deployment: Fiber is pulled or blown through buried HDPE (High-Density Polyethylene) conduits for maximum protection against environmental hazards.

6.2.3 Indoor Termination

The drop cable passes into the home and terminates inside an Access Terminal Box (ATB) or Fiber Rosette on the wall. A highly flexible, bend-insensitive patch cord (typically G.657 standard) connects the ATB to the ONT, preventing signal loss around tight corners.

6.3 Splicing Methods Comparison

Feature	Fusion Splicing	Mechanical / Fast Connectors
Mechanism	Uses a high-voltage electric arc to melt and fuse glass tips permanently.	Physically aligns two fiber ends inside a sleeve using index-matching gel.
Signal Loss	Extremely low (< 0.05 dB to 0.1 dB).	Slightly higher (0.2 dB to 0.5 dB).
Pros & Cons	Most reliable, but requires an expensive fusion splicer machine.	Fast setup, low initial tool cost. Ideal for home drop termination.

Table 6.2: Comparison of Fusion and Mechanical Splicing

6.4 Link Budget & Testing Benchmarks

To prevent collisions on the single-core fiber, different wavelengths are used for upload and download:

6.4.1 Wavelength Multiplexing

- 1490 nm: Downstream data/voice traffic (OLT to ONT).
- 1310 nm: Upstream data upload traffic (ONT to OLT).
- 1550 nm: Optional downstream video/TV overlay.

6.4.2 Testing Parameters

- Optical Power Meter (OPM): Measures absolute light strength at the subscriber end. Acceptable operational range is between -15 dBm and -25 dBm. Readings below -27 dBm constitute an under-power fault.
- OTDR: Measures backscattered light to map the entire physical link and locate precise coordinates of cuts or defects.

6.5 Troubleshooting & Indicators

ONT faceplate status LEDs provide primary diagnostic information during service calls:

ONT Indicator	Status State	Meaning / Corrective Action
LOS (Red)	Steady Glow	Loss of Signal: Total physical fiber cut or severe macro-bend. Check outside plant with OTDR.
PON (Green)	Blinking / Flashing	Registering: Authentication in progress. If persistent, light power is weak or configuration profile is missing.
PON (Green)	Steady Solid Glow	Connected: Symmetrical operational link established successfully.
Power Levels	Below -27 dBm	Power Fault: Dirty connector faces or high-loss splice. Rework connection or clean with a OneClick cleaner.

Table 6.3: ONT Troubleshooting Matrix

Chapter 7

Field Visit: BSNL Exchange Office, Manjeri

7.1 Introduction to the Field Visit

A key part of our training was a technical visit to the BSNL exchange office at Manjeri. The visit allowed us to bridge classroom theory with large-scale telecom practices. This excursion let us trace the physical flow of data from central carrier routers to the local distribution loops.

7.2 Observations and Core Infrastructure

We inspected the active OLT chassis, patch panels, and backup battery systems. The staff explained the process of multiplexing and routing signals into local distribution loops. The OLT works in tandem with key backend routing and authentication nodes: The Broadband Network Gateway (BNG) manages subscriber sessions, allocates dynamic/static IP addresses, and controls user bandwidth. The MNGN (Multiplay Converged packet access Networks) is a secure, carrier-grade packet transmission network deployed within the local exchange. It supports services such as FTTH broadband, leased lines, and enterprise connections over a common network infrastructure. Finally, the OLT transmits these combined services to customer ONTs over fiber loops.

7.3 Conclusion of the Visit

In conclusion, the BSNL exchange visit offered indispensable practical insights. Seeing carrier-grade hardware helped us understand the scale of fiber networks and the complexity of modern broadband delivery.

Part Conclusion: Fiber Optic Infrastructure

In summary, this part of the training detailed the physics and engineering behind high-speed optical communications. We studied light propagation in single-mode cores and analyzed fiber structure and attenuation in distribution cables. Our study of optical splitters and link budget calculations showed how networks maintain signal strength for thousands of users. Through practical field training under Bfone Fibernet, standard industry protocols—including precision fusion splicing, standard color coding sequence alignment, and advanced OTDR-based diagnostic fault-finding—were thoroughly mastered. The BSNL exchange visit clarified wide-area network design, showing how OLTs, BNG nodes, and customer ONTs communicate. Ultimately, this module confirmed that efficient FTTH architectures require a precise balance between strict engineering mathematics and rigorous field deployment standards to maintain continuous, high-performance gigabit connectivity.

Part II

Electrical Home Appliances Maintenance and Troubleshooting

Part Introduction: Domestic Electrical Engineering

Electrical appliances form the core operational infrastructure of the modern home. Devices utilizing resistive heating, induction cooking, or single-phase induction motors apply core principles of thermodynamics and electromagnetism. This section of the report documents the technical details, troubleshooting steps, and maintenance practices studied during our training. The objective of this module is to move beyond simple component replacement to establish a thorough engineering understanding of circuit safety and appliance operations. We discuss basic circuit laws, household wiring layouts, and safety hardware including grounding, circuit breakers (MCBs), and leakage breakers (RCCBs). Following this safety framework, the module provides an isolated component breakdown, operational analysis, and troubleshooting matrix for seven core appliances: the electric iron, induction cooker, mixer grinder, storage water geyser, ceiling fan, wall fan, and centrifugal water pump. Finally, this module presents practical math problems on electricity auditing and electrical charge calculations.

Chapter 8

Fundamentals of Electrical Systems & Safety

8.1 Introduction

Modern homes depend on a variety of electrical appliances. Because these systems run daily, they eventually suffer from bearing wear, insulation aging, and wiring faults. This module was completed to apply electrical theory to practical repair. The course covered safety protocols, wiring layouts, and the step-by-step repair of seven core home appliances. Our goal was to analyze the physics of these appliances—like resistive heating, electromagnetism, and motor action—while practicing safe work habits.

8.2 Fundamentals of Electrical Systems & Safety Overview

A clear understanding of circuit principles and safety gear was established before handling live circuits:

8.2.1 Circuit Behaviors & Laws

- Ohm's Law & Power Equation: Diagnostics rely on Ohm's Law ($V = IR$) and the Power formula ($P = VI = I^2R$). These allow us to calculate expected resistance and identify open or short circuits.
- Series Connections: Current remains constant; voltage divides. Used in safety switches, fuses, and thermostats.
- Parallel Connections: Voltage remains constant; current divides. Used in domestic house wiring so that the failure of one appliance does not shut off others.

$$V = IR \quad \text{and} \quad P = VI = I^2R \quad (8.1)$$

8.2.2 Earthing and Protection Systems

- Earthing (Grounding): Connects the metallic chassis of an appliance directly to the earth. In case of insulation failure where a live wire touches the metal body, current diverts safely to the ground instead of passing through a user.

- Fuse / MCB (Miniature Circuit Breaker): Protects the circuit from overcurrent and short circuits by breaking the circuit when current exceeds safe thresholds.
- ELCB / RCCB (Residual Current Circuit Breaker): Detects minute current leakages (as low as 30mA) escaping to ground by comparing hot and neutral currents, tripping instantly if an imbalance is found.

Chapter 9

Technical Breakdown & Diagnostics of Core Home Appliances

9.1 Electric Iron

A. Working Principle

An electric iron utilizes resistive heating (Joule's Effect). Passing current through a high-resistance heating element converts electrical energy into heat ($H = I^2Rt$). A bimetallic strip thermostat controls the heat. Since the two bonded metals expand at different rates, heating bends the strip to open contacts and cut the circuit. Cooling resets the strip, restarting the heating cycle.

$$H = I^2Rt \quad (9.1)$$

B. Key Components and Parts

- Soleplate: The heavy, polished aluminum or cast-iron base that transfers heat directly to the clothes.
- Heating Element: A ribbon of Nichrome wound around a mica sheet insulated to prevent electric shock while maximizing heat transfer.
- Bimetallic Thermostat: The temperature control mechanism equipped with an adjustable knob to set specific heat thresholds.
- Thermal Fuse: A backup safety device that permanently melts and opens the circuit if the thermostat fails.

C. Appliance Conclusion

The iron diagnostics clarified resistive heating and bimetallic thermostat feedback. Testing showed the importance of secondary thermal fuses for safety.

9.2 Induction Cooker

A. Working Principle

Induction cooking replaces thermal conduction with electromagnetic induction and localized eddy currents. An internal inverter converts 50Hz AC mains into high-frequency AC (20-40kHz). This current flows through an induction coil to generate an alternating magnetic field. Placing a ferromagnetic pan on the glass induces eddy currents in the pan's base. The pan's electrical resistance causes it to heat up directly, leaving the glass top relatively cool. **B. Key Components and Parts**

- Induction Coil: A flat, spirally wound copper coil that generates the alternating magnetic field.
- IGBT (Insulated Gate Bipolar Transistor): The high-speed electronic switch that modulates power and creates the high-frequency current.
- Main PCB & Microcontroller: The 'brain' that monitors user inputs, manages power delivery, and reads sensor data.
- Cooling Fan & RTD (Resistance Temperature Detector): Critical elements for heat dissipation and monitoring pan temperatures to prevent dry boiling.

C. Appliance Conclusion

Diagnosing induction cookers provided valuable knowledge of high-frequency power electronics. It showed how magnetic fields can be used for clean heating and highlighted PCB-level testing.

9.3 Mixer Grinder

A. Working Principle

The mixer grinder is powered by an AC Universal Motor. This series-wound motor operates on both AC and DC. Current passes through the stator and rotor coils via carbon brushes, creating opposing magnetic fields. This generates the high torque and speeds (up to 20,000 RPM) needed to grind and blend food ingredients. **B. Key Components and Parts**

- Stator & Rotor (Armature): The fixed and rotating electromagnetic components that generate mechanical torque.
- Commutator & Carbon Brushes: The assembly that transfers electrical current from the stationary wires to the rotating armature.

9.3.1 Ceiling Fan

A. Working Principle

The ceiling fan uses a Single-Phase Permanent Split Capacitor Induction Motor. A single-phase source lacks a starting torque, so the motor uses starting and running windings. Connecting

a capacitor in series with the starting winding creates a 90-degree phase shift. This simulates a rotating magnetic field, inducing torque in the rotor to rotate the fan. **B. Key Components and Parts**

- Stator & Rotor: Stationary inner core with starting/running windings and the rotating outer casing.
- Run/Start Capacitor: Typically 2.25 μ F to 2.5 μ F capacitor used to split the phase and start rotation.
- Ball Bearings: Precision bearings that reduce mechanical friction, ensuring smooth and quiet operation.
- Down-Rod Assembly: Structural hardware used to safely suspend the fan from the ceiling.

C. Appliance Conclusion

Ceiling fan repair sessions demonstrated phase separation in single-phase induction. We learned how to isolate electrical winding faults from mechanical bearing seizure.

9.3.2 Wall Fan

A. Working Principle

The wall fan runs on a single-phase capacitor motor, incorporating swiveling oscillation. The motor turns the blades while a gear train or secondary micro-motor sweeps the fan head. Speed adjustment is achieved by selecting different winding taps or through a TRIAC speed controller on the PCB. **B. Key Components and Parts**

- Fan Blades and Guard: Aerodynamically pitched plastic or metal blades protected by a wire mesh safety cage.
- Oscillation Gear Assembly / Clutch Knob: A mechanical gear train connected to the rotor shaft that translates high-speed rotation into slow, wide-angle swiveling motion.
- Speed Control Switch (Pull-Cord / Push-Button / Remote PCB): Mechanisms to toggle between different motor speed taps or control electronic voltage regulation.
- Thermal Cut-Off (TCO): A compact thermal fuse embedded directly inside the motor windings to protect the motor from burning out if the blades get obstructed or jammed.

C. Appliance Conclusion

The wall fan provided an study on mechanical linkages combined with induction motors. Diagnostics focused on winding thermal fuses and gear alignment issues.

Chapter 10

Troubleshooting & Maintenance Practice Matrix

ONT Indicator	Status State	Meaning / Corrective Action
LOS (Red)	Steady Glow	Loss of Signal: Total physical fiber cut or severe macro-bend. Check outside plant with OTDR.
PON (Green)	Blinking / Flashing	Registering: Authentication in progress. If persistent, light power is weak or configuration profile is missing.
PON (Green)	Steady Solid Glow	Connected: Symmetrical operational link established successfully.
Power Levels	Below -27 dBm	Power Fault: Dirty connector faces or high-loss splice. Rework connection or clean with a OneClick cleaner.

Table 10.1: ONT Troubleshooting Matrix Summary

Appliance	Common Symptom	Root Cause Detected	Corrective Action Performed
Electric Iron	No heating; indicator light off	Open circuit in thermal fuse / Broken power cord	Replaced thermal fuse; trimmed and re-terminated power cord.
Induction Cooker	Dead display; won't power up	Blown main fuse due to shorted IGBT	Replaced the shorted IGBT and main glass fuse; checked gate driver circuit.
Mixer Grinder	Sparks from bottom; sudden stop	Worn carbon brushes / Overload switch tripped	Replaced carbon brushes; cleared jar jam and reset the bottom OLP switch.
Water Heater	Water not heating; indicator on	Scale buildup or open-circuit in heating element	Drained tank, removed limescale, and replaced the immersion heating element.
Ceiling Fan	Humming sound; spins very slowly	Degraded or leaked capacitor	Desoldered/disconnected old capacitor, verified value, and installed new 2.5 μ F unit.
Wall Fan	Fan rotates but does not oscillate / Overheating	Stripped linkage gears / Dry rotor shaft	Replaced the plastic gear wheel assembly; cleaned and lubricated the rotor shaft with machine oil.
Pump Motor	Motor hums but does not rotate	Seized bearings / Defective starter capacitor	Freed impeller manually; replaced bearings and capacitor.

Table 10.2: Appliance Troubleshooting & Maintenance Practice Matrix

10.1 Internship Outcomes & Skills Acquired

- **Safety Instinct Diagnostics:** Developed a strict protocol-first mindset—always verifying isolation with a neon tester, utilizing insulation mats, and ensuring appropriate grounding before applying mains voltage.
- **Systematic Fault Profiling:** Shifted away from random guesswork to structured troubleshooting: tracing faults from the source point (plug/cord) through protection systems (OLPs/fuses)

to control systems and finally the load.

- Instrument Competency: Attained fluid precision in utilizing digital multimeters to measure resistance drops, AC loop voltages, and verifying open/short circuits.

Chapter 11

Applied Practical Calculations

11.1 Domestic Electricity Consumption Auditing

Electricity billing is based on energy consumed in kilowatt-hours (kWh), where 1 Unit represents 1 kWh.

Find the total energy consumed in a 30-day month by a home running two 40W tube lights for 6 hours a day, three 65W ceiling fans for 8 hours a day, and other appliances drawing 150W for 2 hours daily.

$$E_{\text{Lights}} = 2 \times 40 \text{ W} \times 6 \text{ hrs/day} \times 30 \text{ days} = 14,400 \text{ Wh} = 14.4 \text{ kWh}$$

$$E_{\text{Fans}} = 3 \times 65 \text{ W} \times 8 \text{ hrs/day} \times 30 \text{ days} = 46,800 \text{ Wh} = 46.8 \text{ kWh}$$

$$E_{\text{Misc}} = 150 \text{ W} \times 2 \text{ hrs/day} \times 30 \text{ days} = 9,000 \text{ Wh} = 9.0 \text{ kWh}$$

$$E_{\text{Total}} = 14.4 \text{ kWh} + 46.8 \text{ kWh} + 9.0 \text{ kWh} = 70.2 \text{ kWh (Units)}$$

11.2 Charge Calculation in Audio Systems

Determine the total charge transported through a 230V electrical system that consumes 8 MJ of energy during its operating cycle.

$$\text{Given: } V = 230 \text{ V, } E = 8 \times 10^6 \text{ J}$$

$$\text{Equation: } E = qV$$

$$\begin{aligned} q &= \frac{E}{V} \\ &= \frac{8 \times 10^6 \text{ J}}{230 \text{ V}} \\ &= 3.48 \times 10^4 \text{ Coulombs (C)} \end{aligned}$$

Part Conclusion: Domestic Electrical Engineering

To conclude, this second part linked academic physics concepts with practical appliance diagnostics. Studying the components and wiring of seven core appliances provided valuable insight into domestic electrical layouts. We learned to trace faults, such as open bimetallic contacts in irons or degraded start capacitors in fans, using multimeters. The successful implementation of practical testing logic highlighted how embedded safety devices like thermal cut-offs (TCOs), overload protectors (OLPs), and dual-stage pressure relief valves are critical for preventing motor burnouts and protecting users. Additionally, completing energy audits and charge calculations showed how to monitor power consumption. Ultimately, the mechanical repair skills and systemic troubleshooting methods gained during this maintenance module form a critical practical toolkit, ensuring that home environments remain completely safe and operationally productive.